DS8001: Design of Algorithms and Programming for Massive Data

Graduate Program in Data Science and Analytics
Toronto Metropolitan University

Fall 2026

Syllabus

Instructor: Prof. Mucahit Cevik
Office location: EPH 416
Office hour: Wednesday 4-5pm (ET), via Zoom
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TAs: TBD

Course web page: D2L (my.torontomu.ca)

Discussion board: D2L

P.S. Rather than emailing the teaching staff, please post your questions on D2L

NOTE: In accordance with the Policy on TMU Student E-mail Accounts (Policy 157), TMU requires that any electronic communication by students to TMU faculty or staff be sent from their official TMU email account

Lectures: Thursday 6.10pm - 9.00pm (ET)
via Zoom (see Announcements in D2L for zoom link)

Compulsory text: None

Supplementary texts:
- ◇ *Big Data: Algorithms, Analytics, and Applications*, CRC Press, 2015, by K.-C Li, H. Jiang, L. T. Yang, A. Cuzzocrea, (eds.) Price (link): $USD 62.99
- ◇ *Algorithm Design,* Pearson, 2006, by J. Kleinberg and E. Tardos Price (link): $USD 89.94
- ◇ *Algorithms*, 2019, by J. Erickson Price (link): Free
- ◇ *Mining of Massive Datasets*, 2020, by J. Leskovec, A. Rajaraman, J. Ullman Price (link): Free

Course information

Official course description

To introduce students to the theory and design of algorithms to acquire and process (sorting, analyzing) large dimensional data. Advanced data structures, graph algorithms, and algebraic algorithms. Complexity analysis, complexity classes, and NP Completeness. Introduction to information retrieval and streaming algorithms. Big data applications.

Prerequisites/co-requisites: None

Course objectives and intended learning outcomes

At the end of this course, the successful student will be able to:

1. Analyse the problems, algorithms and identify the complexities of solving them
2. Identify existing or design new algorithms to process massive data

Topics to be covered

- Divide and Conquer
- Greedy Algorithms
- Graphs
- NP Complexity
- Dynamic Programming
- Algorithms for Unconstrained Optimization
- Search Algorithms
- Scalable Indexing for Big Data Processing, Hashing
- Indexing for Similarity Search for Large Data Sets, Finding Similar Items
- Recommender Systems
- Massively Parallel Algorithms

Teaching methodology

This is a lecture-based course in which the students are encouraged to apply the techniques learned in the classroom through computer lab exercises.

Grading

The course grade will be based on a weighted average of the following components:

No.	Title	Individual/group	Value	Detail/general description
1.	Assignments	Group (1-2 students)	16%	Two assignments in total
2.	Lab Exercises	Individual	5%	8 exercises, lowest 3 scores will be dropped. Lab exercises will be released and submitted through D2L; dates will be posted on D2L.
3.	Exam	Individual	30%	Take home exam, open book/notes. Exam will only have written questions. Exam duration is ~ 3 **hours**. Exam will be released/collected through D2L.
4.	Quiz $\times$ 2	Individual	14%	Online, duration: $\approx$ **30-40 minutes**.
5.	Project	Group (1-2 students)	35%	See project guidelines

Note: You can object to your grade for a specific grading item **only** within two weeks after the grade is announced. The objection for any coding question must be accompanied by a pdf file (e.g., as converted from word/google docs) that details why your submitted code justifies more credit. You must report whether your code generates output, and how the input/output follows or deviates from the sample input/output.

Late work submission

Late submissions will not be accepted except where academic consideration or accommodation applies.

Feedback and grades

First two assignment grades will be announced before the course drop deadline.

Tentative schedule

Following is a list of general topics which we will cover in this course, and a tentative schedule. Topics, number of assignments, and dates are subject to change. If time permits we may cover some topics not explicitly listed here. For the up-to-date version (of all the material, e.g., syllabus, lecture slides, assignments and their solutions, tutorial material), please check D2L regularly.

Week #	**Topic**	**Readings**[†]	**Notes**
1 (Sep 10)	Introduction to Algorithms	LN	
2 (Sep 17)	Divide and Conquer	Ch5 (Kleinberg & Tardos)	
3 (Sep 24)	Greedy Algorithms, Graph Applications	Ch4 (Kleinberg & Tardos)	HW1 out (Sep 24, 9pm)
4 (Oct 1)	Dynamic Programming	Ch6 (Kleinberg & Tardos)	
5 (Oct 8)	Algorithms for Unconstrained Optimization	Ch9 (Boyd & Vandenberghe)	HW1 due (Oct 8, 6pm), HW2 out (Oct 8, 9pm)
6 (Oct 15)	—	—	Fall Break
7 (Oct 22)	Search Algorithms	Ch3-4 (Norvig & Russell)	Quiz - 1
8 (Oct 29)	Recommender Systems	Ch9 (Leskovec et al)	HW2 due (Oct 26, 6pm)
9 (Nov 5)	Scalable Indexing, Hashing	Ch1 (Li et al)	PROJECT PROPOSAL DUE (Nov 5, 6pm)
10 (Nov 12)	NP Complexity	Ch8 (Kleinberg & Tardos)	Quiz - 2
11 (Nov 19)	—	—	EXAM (take home) (Nov 19, 6pm - 9pm)
12 (Nov 26)	Indexing for Similarity Search, Finding Similar Items	Ch3 (Li et al), Ch3 (Leskovec et al)	
13 (Dec 3)	Massively Parallel Algorithms	LN	
Dec 7	—	—	PROJECT DUE (tentative) (Dec 7, 6pm)

[†] These are *recommended* readings. Citations are as follows. LN: Lecture notes, Kleinberg & Tardos: Kleinberg and Tardos (2006), Leskovec et al: (Leskovec et al., 2020), Boyd & Vandenberghe: Boyd and Vandenberghe (2004), Norvig & Russell: Norvig and Russell (2003), Li et al: Li et al. (2015) – see references for readings

References

S. P. Boyd and L. Vandenberghe. *Convex optimization*. Cambridge university press, 2004.

J. Kleinberg and E. Tardos. *Algorithm Design*. Pearson Education India, 2006.

J. Leskovec, A. Rajaraman, and J. D. Ullman. *Mining of Massive Datasets*. Cambridge University Press, 2020.

K.-C. Li, H. Jiang, L. T. Yang, and A. Cuzzocrea. *Big data: Algorithms, analytics, and applications*. CRC Press, 2015.

P. Norvig and S. J. Russell. Artificial intelligence a modern approach, 2003.

University Academic Policies

Students are reminded that they are required to adhere to all relevant university policies found in their online course shell in D2L and/or on the Senate website.

For information on academic policies pertaining to issues such as course management, grading practices, and appeals, students are to refer to the TMU Senate Policies: Policy 170(b) (Formerly #164) – Graduate Status, Enrolment, and Evaluation, Policy 166 - Course Management, and Policy 167 - Academic Consideration.

Academic Integrity

TMU's Policy 60 - Academic Integrity Policy, applies to all students at the University. Note however, that expectations for **graduate students** are set to higher levels. If there is a finding of academic misconduct, **the minimum penalty for misconduct with respect to work submitted in a course by a graduate student is a grade of "zero" (0) on the work**. Graduate students who are found to have committed academic misconduct may also have a Disciplinary Notation (DN) placed on their academic record, **which will exclude them to be eligible for any scholarships and/or awards**. In addition, they may be assigned one or more of the penalties ranging from **a grade of "F" in the course, to DA (Disciplinary action), DA-S (Disciplinary action with suspension), (DW) Disciplinary withdrawal, up to an expulsion or even revocation of a degree**.

Forms of academic misconduct include plagiarism, cheating, supplying false information to the University, and other acts. The most common form of academic misconduct is plagiarism. Plagiarism is a serious academic offence, and penalties can be severe. In any academic exercise, plagiarism occurs when one offers as one's own work the words, data, ideas, arguments, calculations, designs, or productions of another without appropriate attribution or when one allows one's work to be copied.

Furthermore, the unauthorized use of intellectual property of others, including your professor, for distribution, sale, or profit is expressly prohibited. Intellectual property includes, but is not limited to: slides, lecture notes, presentation materials used in and outside of class, lab manuals, course packs, exams, etc.

All academic work must be submitted using the citation style approved by the instructor. Students may refer to the TMU Library's list of Citations and Style Guides for more information.

It is assumed that all examinations and work submitted for evaluation and course credit will be the product of individual effort, except in the case of group projects arranged for and approved by the course instructor. Submitting the same work to more than one course, without instructor approval, is also considered a form of plagiarism.

For more detailed information on these issues, please refer to the full online text for the Toronto Metropolitan University Senate Policy 60: Academic Integrity. For more information on how to avoid academic misconduct situations, for clues and tips, visit the Academic Integrity website.

Generative AI Course Policy

Generative AI tools (e.g. ChatGPT, Gemini, Grammarly, Perplexity, DeepL Translator) may be used only to correct grammar and language in text authored by the student; they may not be used to generate or substantially rewrite answers, code, analyses, or other assessed content. Failure to stay within these limits will be considered a breach of Policy 60.

Turnitin or another originality detection software

Turnitin is a plagiarism prevention and detection service to which TMU subscribes. It is a tool to assist instructors in determining the similarity between students' work and the work of other students who have submitted papers to the site (at any university), internet sources, and a wide range of books, journals and other publications. While it does not contain all possible sources, it gives instructors some assurance that

students' work is their own. No decisions are made by the service; it generates an "originality report," which instructors must evaluate to judge if something is plagiarized. Students agree by taking this course that their written work will be subject to submission for textual similarity review to Turnitin. Instructors can opt to have student's papers included in the Turnitin database or not. Use of the Turnitin service is subject to the terms-of-use agreement posted on the Turnitin website. Students who do not want their work submitted to this plagiarism detection service must, by the end of the second week of class, consult with their instructor to make alternate arrangements. Students who choose not to have their papers screened for textual similarity review by Turnitin may be required to submit additional work with their assignment/project report. For example:

- an annotated bibliography of each source used in your paper; and/or
- the first few pages of each cited source used in your paper

Even when an instructor has not indicated that a plagiarism detection service will be used, or when a student has opted out of the plagiarism detection service, if the instructor has reason to suspect that an individual piece of work has been plagiarized, the instructor is permitted to submit that work in a non-identifying way to any plagiarism detection service.

Important Resources Available at Toronto Metropolitan University

- The University Libraries provide research workshops and individual consultation appointments. There is a drop-in Research Help desk on the second floor of the library, and students can use the Library's virtual research help service to speak with a librarian, or book an appointment to meet in person or online.
- Student Life and Learning Support offers group-based and individual help with writing, math, study skills, and transition support, as well as resources and checklists to support students as online learners.
- You can submit an Academic Consideration Request when an extenuating circumstance has occurred that has significantly impacted your ability to fulfill an academic requirement. You may always visit the Senate website and select the blue radio button on the top right hand side entitled: Academic Consideration Request (ACR) to submit this request.

  For Extenuating Circumstances, Policy 167: Academic Consideration allows for a once per semester ACR request without supporting documentation if the absence is less than 3 days in duration and is not for a final exam/final assessment. Absences more than 3 days in duration and those that involve a final exam/final assessment, always require documentation. Students must notify their instructor once a request for academic consideration is submitted. See Senate Policy 167: Academic Consideration.
- Information on Copyright for Faculty and students.
- Information on Academic Integrity for Faculty and students.

Academic Accommodation Support

Academic Accommodation Support (AAS) is the university's disability services office. AAS works directly with incoming and returning students looking for help with their academic accommodations. AAS works with any student who requires academic accommodation regardless of program or course load.

- Learn more about Academic Accommodation Support.
- Learn how to register with AAS.
- Learn about Policy 159: Academic Accommodation of Students with Disabilities

Academic Accommodations (for students with disabilities) and Academic Consideration (for students faced with extenuating circumstances that can include short-term health issues) are governed by two different university policies. Learn more about Academic Accommodations versus Academic Consideration and how to access each.

Wellbeing Support

At Toronto Metropolitan University, we recognize that things can come up throughout the term that may interfere with a student's ability to succeed in their coursework. These circumstances are outside of one's control and can have a serious impact on physical and mental well-being. Seeking help can be a challenge, especially in those times of crisis.

If you are experiencing a mental health crisis, please call 911 and go to the nearest hospital emergency room. You can also access these outside resources at any time:

- Distress Line: 24/7 line for if you are in crisis, feeling suicidal or in need of emotional support (phone: 416–408–4357)
- Good2Talk: 24/7-hour line for postsecondary students (phone: 1-866-925-5454)
- Keep.meSAFE: 24/7 access to confidential support through counsellors via My SSP app or 1-844-451-9700

If non-crisis support is needed, you can access these campus resources:

- Centre for Student Development and Counselling: 416-979-5195 or email csdc@torontomu.ca
- Consent Comes First – Office of Sexual Violence Support and Education: 416-919-5000 ext 3596 or email osvse@torontomu.ca
- Medical Centre: call (416) 979-5070 to book an appointment

We encourage all Toronto Metropolitan University community members to access available resources to ensure support is reachable. You can find more resources available through the Toronto Metropolitan University Mental Health and Wellbeing website.